

SAFETY DATA SHEET

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier:

Ferrous fumarate

Chemical name: Ferrous fumarate

CAS number: 141-01-5

EC number: 205-447-7

Registration number is not available for this substance since this substance or its use is exempted from registration based on Article 2 of REACH regulation, or registration is not necessary due to the annual tonnage band, or the registration is scheduled to a later registration date.

1.2. Relevant identified uses of the substance or mixture and uses advised against:

Fumarate is manufactured for export to the pharmaceutical industry in pharmaceutical grade quality. For industrial use.

1.3. Details of the supplier of the safety data sheet:

Information about the manufacturer:

FerroPharma Kft.

6766 Dóc, Erdősor 13.

Hungary

Tel: +36-62-570-010

1.3.1. Responsible person:

Sebestyén Tibor

E-mail:

tibor.sebestyen@ferropharma.hu

1.4. Emergency telephone number:

Please fill in

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture:

Classification according to Regulation (EC) No 1272/2008 (CLP):

Not considered as hazardous substance.

Hazard statements: No hazard statements.

2.2. Label elements:

Chemical name: Ferrous fumarate

CAS number: 141-01-5

EC number: 205-447-7

Hazard statements: No hazard statements.

Precautionary statements: No precautionary statements.

2.3. Other hazards:

Contact with eyes may cause mechanical irritation.

The enrichment of dry fine dust leads to the danger of dust explosion.

Results of PBT and vPvB assessment: Not applicable.

Endocrine disrupting property: The substance is not considered to have endocrine disrupting properties.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances:

Chemical name: Ferrous fumarate; (Iron(II) fumarate)
CAS number: 141-01-5
EC number: 205-447-7
Molecular formula: $C_4H_2FeO_4$

Chemical characterization:

	Complies to current version of EP	Complies to current version of USP	Complies to current version of FCC
Assay	Min. 93.0 – Max. 101.0 %	97.0-101.0 %	97.0-101.0 %
Ferric ion (Fe^{3+})	Max 2.0 %	NMT 2 %	NMT 2 %
Loss on drying	Max 1.0 %	NMT 1.5 %	NMT 1.5 %
Sulphate	Max 0.2 %	NMT 0.2 %	NMT 0.2 %
Arsenic	Max 1,5 ppm	NMT 3 ppm	-----
Mercury	-----	NMT 3 ppm	NMT 3 ppm
Chromium	Max 200 ppm	-----	-----
Lead	Max 3 ppm	NMT 10 ppm	NMT 2 ppm
Cadmium	-----	-----	-----
Nickel	Max 50 ppm	-----	-----
Zinc	Max 500 ppm	-----	-----
Cobalt	Max 20 ppm	-----	-----
Vanadium	Max 40 ppm	-----	-----

The value of each component complies with the effective USP, FCC and EP.

The above characterization is not specification. For exact heavy metal content see the specification.

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures:

General information: The product does not cause permanent injury. In case of doubt or persistent symptoms, seek medical advice.

INGESTION:

Measures:

- Drink plenty of water.
- If suspected symptoms appear, seek medical advice.

INHALATION:

Measures:

- Take the victim into fresh air.
- Get medical attention for any breathing difficulty.

SKIN CONTACT:

Measures:

- The product does not irritate the skin.

EYE CONTACT:

Measures:

- In case of contact with eyes, flush with water holding eyelids apart for several minutes.
- In case of persistent symptoms, obtain medical help.

4.2. Most important symptoms and effects, both acute and delayed:

After swallowing: if symptoms persist, consult doctor.

4.3. Indication of any immediate medical attention and special treatment needed:

No special treatment needed; treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

- 5.1. **Extinguishing media:**
5.1.1. **Suitable extinguishing media:**
CO₂, powder or water spray. Fight larger fires with plenty of water or alcohol resistant foam
5.1.2. **Unsuitable extinguishing media:**
No data available.
- 5.2. **Special hazards arising from the substance or mixture:**
In case of fire the product does not burn, but glows.
In case of fire, carbon-monoxide and carbon-dioxide can be released.
- 5.3. **Advice for firefighters:**
Wear full protective clothing and self-contained breathing apparatus.
Avoid inhalation of explosion gases or combustion gases.

SECTION 6: ACCIDENTAL RELEASE MEASURES

- 6.1. **Personal precautions, protective equipment and emergency procedures:**
6.1.1. **For non-emergency personnel:**
Allow only well-trained experts wearing suitable protective clothing to abide in the field of accident.
6.1.2. **For emergency responders:**
No special precautions required.
- 6.2. **Environmental precautions:**
Dispose of the spillage and the resulting waste according to the applicable environmental regulations. Do not allow the product and the resulting waste to enter sewers/soil/surface or ground water. Notify the respective authorities in accordance with local law in the case of environmental pollution immediately.
- 6.3. **Methods and material for containment and cleaning up:**
Spilled product should be collected mechanically or with vacuum-cleaner to avoid the inhalation of dust.
- 6.4. **Reference to other sections:**
For further and detailed information see Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

- 7.1. **Precautions for safe handling:**
Observe conventional hygiene precautions.
Prevent formation of dust.
Wash your hand with plenty of water before eating, drinking or smoking.
Clean the dust of the product from your hand and clothes after leaving the store, where the product is used or stored.
- Technical measures:**
In case of proper use, special measures are not necessary.
- Precautions against fire and explosion:**
No special measures required.
- 7.2. **Conditions for safe storage, including any incompatibilities:**
Technical measures and storage condition:
Keep in the original, well closed container, away from oxidizing agents.
Eating, drinking and smoking is not allowed where the product is stored.
Protect from exposure to the light.
Incompatible materials: See Section 10.5.
Packaging material: No special prescriptions.
- 7.3. **Specific end use(s):**
No specific instructions available.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

- 8.1. **Control parameters:**

Occupational exposure limit values (Commission Directive (EC) No 2000/39 of 8 June 2000):
The substance is not regulated with exposure limit value.

DNEL values		Oral exposure		Dermal exposure		Inhalative exposure	
		Short term (acute)	Long term (chronic)	Short term (acute)	Long term (chronic)	Short term (acute)	Long term (chronic)
Consumer	Local	no data	no data	no data	no data	no data	no data
	Systemic	no data	no data	no data	no data	no data	no data
Worker	Local	no data	no data	no data	no data	no data	no data
	Systemic	no data	no data	no data	no data	no data	no data

PNEC values		
Compartment	Value	Note(s)
Freshwater	no data	no notes
Marine water	no data	no notes
Freshwater sediment	no data	no notes
Marine water sediment	no data	no notes
Sewage Treatment Plant (STP)	no data	no notes
Intermittent release	no data	no notes
Secondary poisoning	no data	no notes
Soil	no data	no notes

8.2. Exposure controls:

In case of a hazardous material with no controlled concentration limit it is the employer's duty to keep concentration levels down to a minimum achievable by existing scientific and technological means, where the hazardous substance poses no harm to workers.

8.2.1. Appropriate engineering controls:

In pursuance of work is proper foresight needed to avoid leaking onto clothes and floors and to avoid contact with eyes and skin. For further information see Section 7.

8.2.2. Individual protection measures, such as personal protective equipment:

Use the product only in closed place. Wash hands before breaks and at the end of the work.

Observe the general safety regulations when handling chemicals.

- Eye/face protection:** In case of contact with the product or the product dust use appropriate protective glasses (EN ISO 16321-1:2022; EN 166).
- Skin protection:**
 - Hand protection:** Use appropriate protective gloves (EN 374).
 The glove material should be impermeable and resistant to the product.
 Due to missing tests no recommendation to the glove material can be given for the product/chemical mixture.
 Select glove material based on the penetration time, rates of diffusion and degradation.
 The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.
 - Other:** Use appropriate protective clothing.
- Respiratory protection:** Suitable respiratory equipment with fine particle filter and sufficient ventilation is necessary (EN 143; Recommended Filter type: Filter P 1).
- Thermal hazards:** No thermal hazards known.

8.2.3. Environmental exposure controls:

No specific prescription.

The requirements detailed in Section 8 assume skilled work under normal conditions and usage of the product for appropriate aims. If conditions differ from normal or work is carried out under extreme conditions, an expert's advice is necessary before deciding upon further protective measures.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties:

Parameter	Value / Test method / Remarks
1. Physical state	fine, stable homogeneous powder
2. Colour	from reddish orange to reddish-brown
3. Odour, odour threshold	odourless or almost odourless
4. Melting point/freezing point	> 280°C
5. Boiling point or initial boiling point and boiling range	no data*
6. Flammability	no data*
7. Lower and upper explosion limit	no data*
8. Flash point	not applicable
9. Auto-ignition temperature	no data*
10. Decomposition temperature	no data*
11. pH	no data*
12. Kinematic viscosity	no data*
13. Solubility in water in other solvents	slightly soluble very slightly soluble in ethanol
14. Partition coefficient n-octanol/water (log value)	no data*
15. Vapour pressure	no data*
16. Density and/or relative density	1.75 g/cm ³
17. Relative vapour density	no data*
18. Particle characteristics	no data*

9.2. Other information:

9.2.1. Information with regard to physical hazard classes:

Explosive properties: The enrichment of dry fine dust leads to the danger of dust explosion.
 Ignition temperature: 400°C (Godbert-Greenwald of Ofen)

9.2.2. Other safety characteristics:

No other characteristics available.

*: The manufacturer did not carry out any tests on this parameter for the product or the results of the tests are not available at the time of publication of the data sheet, or the property is not applicable for the product.

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity:

No reactivity known.

10.2. Chemical stability:

Chemically stable at normal temperature, handling and storage condition.

10.3. Possibility of hazardous reactions:

No hazardous reactions known.

10.4. Conditions to avoid:

No conditions to avoid known.

10.5. Incompatible materials:

No incompatible materials known.

10.6. Hazardous decomposition products:

Carbon monoxide and carbon dioxide.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008:

Acute toxicity: Based on available data, the classification criteria are not met.

Skin corrosion/irritation: Based on available data, the classification criteria are not met.

Serious eye damage/irritation: Based on available data, the classification criteria are not met.

Respiratory or skin sensitisation: Based on available data, the classification criteria are not met.

Germ cell mutagenicity: Based on available data, the classification criteria are not met.

Carcinogenicity: Based on available data, the classification criteria are not met.

Reproductive toxicity: Based on available data, the classification criteria are not met.

STOT-single exposure: Based on available data, the classification criteria are not met.

STOT-repeated exposure: Based on available data, the classification criteria are not met.

Aspiration hazard: Based on available data, the classification criteria are not met.

11.1.1. Summaries of the information derived from the test conducted:

No data available.

11.1.2. Relevant toxicological properties:

Information about the product:

Acute toxicity:

LD50 (oral): 3850 mg/kg

11.1.3. Information on likely routes of exposure:

Ingestion, inhalation, skin contact, eye contact.

11.1.4. Symptoms related to the physical, chemical and toxicological characteristics:

Primary irritant effect:

On the skin: no irritant effect.

On the eye: may cause mechanical irritation.

Sensitization: no sensitizing effect known.

11.1.5. Delayed and immediate effects as well as chronic effects from short and long-term exposure:

No data available.

11.1.6. Interactive effects:

No data available.

11.1.7. Absence of specific data:

No information.

11.2. Information on other hazards:

Endocrine disrupting properties:

Endocrine disrupting property: The substance is not considered to have endocrine disrupting properties.

Other information:

Due to experience and product composition, hazardous effects are not possible, if the product is used properly and according to the directions.

The product is listed as food and feed additive and active pharmaceutical ingredient.

The substance is not subject to classification according to the latest version of EU lists.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity:

The substance is not classified as hazardous for the environment.

12.2. Persistence and degradability:

Biodegradable.

12.3. Bioaccumulative potential:

No data available.

12.4. Mobility in soil:

No data available.

12.5. Results of PBT and vPvB assessment:

Not applicable.

12.6. Endocrine disrupting properties:

Endocrine disrupting property: The substance is not considered to have endocrine disrupting properties.

12.7. Other adverse effects:

General notes: slightly hazardous for water.

Undiluted product or large quantities of the product or reach ground water, water course or sewage system is avoided.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods:

Disposal according to the local regulations.

13.1.1. Information regarding the disposal of the product:

Dispose of in accordance with applicable regulations.

Recommendations: small quantity can be disposed as household waste.

The residue of the product is irritant. Handle the used packaging material in accordance with the official local regulations.

List of Waste Code:

No waste disposal key according to the List of Waste Code (LoW code) can be determined for this product, as only the purpose of application defined by the user enables an allocation. The LoW code number has to be determined after a discussion with a waste disposal specialist.

13.1.2. Information regarding the disposal of the packaging:

Dispose of in accordance with applicable regulations.

13.1.3. Physical/chemical properties that may affect waste treatment options shall be specified:

No data available.

13.1.4. Sewage disposal:

No data available.

13.1.5. Special precautions for any recommended waste treatment:

No data available.

SECTION 14: TRANSPORT INFORMATION

ADR/RID; ADN; IMDG; IATA:

Not subject to the conventions of carriage of dangerous goods.

14.1. UN number or ID number:

No UN or ID number.

14.2. UN proper shipping name:

No proper shipping name.

14.3. Transport hazard class(es):

No transport hazard classes.

14.4. Packing group:

No packing group.

14.5. Environmental hazards:

Not applicable.

14.6. Special precautions for user:

No relevant information available.

14.7. Maritime transport in bulk according to IMO instruments:

Not applicable.

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture:

REGULATION (EC) No 1907/2006 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive (EC) No 1999/45 and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive (EEC) No 76/769 and Commission Directives (EEC) No 91/155, (EEC) No 93/67, (EC) No 93/105 and (EC) No 2000/21

REGULATION (EC) No 1272/2008 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives (EEC) No 67/548 and (EC) No 1999/45, and amending Regulation (EC) No 1907/2006

COMMISSION REGULATION (EU) 2020/878 of 18 June 2020 amending Annex II to Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)

15.2. Chemical safety assessment: Has not been carried out.

SECTION 16: OTHER INFORMATION

Information regarding the revision of the safety data sheet: No information.

Literature references / data sources:

Safety data sheet issued by the manufacturer (01. 05. 2023, version 10, EN).

Relevant hazard statements (code and full text) of Sections 2 and 3: No relevant statements.

Training advice: No data available.

Full text of the abbreviations in the safety data sheet:

ADN: European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways.

ADR: Agreement concerning the International Carriage of Dangerous Goods by Road.

ATE: Acute Toxicity Estimate.

AOX: Adsorbable organic halides.

BCF: Bioconcentration factor.

BOD: Biological Oxygen Demand.

CAS number: Chemical Abstract Service number.

CLP: Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures.

CMR effects: Carcinogenic, mutagenic, reprotoxic effects.

COD: Chemical Oxygen Demand.

CSA: Chemical Safety Assessment.

CSR: Chemical Safety Report.

DNEL: Derived-No-Effect-Level.

ECHA: European Chemical Agency.

EC: European Community.

EC number: EINECS and ELINCS numbers (see also EINECS and ELINCS).

EEC: European Economic Community.

EEA: European Economic Area (EU + Iceland, Liechtenstein and Norway).

EINECS: European Inventory of Existing Commercial Chemical Substances.

ELINCS: European List of Notified Chemical Substances.

EN: European Norm.

EU: European Union.

EuPCS: European Product Categorisation System.

EWC: European Waste Catalogue (replaced by LoW – see below).

GHS: Globally Harmonized System of Classification and Labelling of Chemicals.

IATA: International Air Transport Association.

ICAO-TI: Technical Instructions for the Safe Transport of Dangerous Goods by Air.

IMDG: International Maritime Dangerous Goods.

IMO: International Maritime Organization.

IMSBC: International Maritime Solid Bulk Cargoes.

IUCLID: International Uniform Chemical Information Database.

IUPAC: International Union of Pure and Applied Chemistry.

Kow: n-Octanol - Water Partition Coefficient.

LC50: Lethal concentration resulting in 50 % mortality.

LD50: Lethal dose resulting in 50 % mortality (median lethal dose).

LoW: List of Waste.

LOEC: Lowest Observed Effect Concentration.

LOEL: Lowest Observed Effect Level.

NOEC: No Observed Effect Concentration.

NOEL: No Observed Effect Level.

NOAEC: No Observed Adverse Effect Concentration.

NOAEL: No Observed Adverse Effect Level.

OECD: Organization for Economic Cooperation and Development.

OSHA: Occupational Safety and Health Administration.

PBT: Persistent, Bioaccumulative and Toxic.

PNEC: Predicted No Effect Concentration.

QSAR: Quantitative Structure Activity Relationship.

REACH: Regulation 1907/2006/EC concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals.

RID: Regulations Concerning the International Transport of Dangerous Goods by Rail.



SCBA: Self Contained Breathing Apparatus.

SDS: Safety Data Sheet.

STOT: Specific Target Organ Toxicity.

SVHC: Substances of Very High Concern.

UN: United Nations.

UVCB: Chemical Substances of Unknown or Variable Composition, Complex Reaction Products or of Biological Materials.

VOC: Volatile Organic Compound.

vPvB: very Persistent and very Bioaccumulative.

This safety data sheet had been prepared on the basis of information provided by the manufacturer/supplier and conform to the relevant regulations.

The information, data and recommendations contained herein are provided in good faith, obtained from reliable sources and believed to be true and accurate as of the date issued; however, no representation is made as to the comprehensiveness of the information.

The SDS shall be used only as a guide for handling the product; in the course of handling and using the product other considerations may arise or be required.

Users are cautioned to determine the appropriateness and applicability of the above information to their particular circumstances and purposes and assume all risk associated with the use of this product.

It is the responsibility of the user to fully comply with local, national and international regulations concerning the use of this product.

Safety data sheet was prepared by:
MSDS-Europe
International branch of ToxInfo Kft.

Professional help regarding
the explanation of the safety
data sheet:
+36 70 335 8480; info@msds-europe.com
www.msds-europe.com

